

Title

Structured Digital Lifestyle Intervention in the Interpregnancy Period: A Multisite Implementation Study

Abstract

Background: Weight gain between pregnancies and during pregnancy is associated with obstetric and neonatal complications. Few interventions target this critical window.

Methods: This multisite randomized protocol will evaluate a comprehensive lifestyle program delivered primarily through digital platforms in postpartum women at increased risk for adverse outcomes. Women will be recruited in the immediate postpartum period (days 2-3) and randomized at week 6 postpartum. The intervention includes structured coaching sessions delivered in-person or remotely, a Bluetooth-connected mobile application linking to wearable activity trackers and digital scales, and content addressing nutrition, physical activity, weight management, and mental well-being. Intervention will be delivered during the postpartum period (4 sessions over 6 months) and again during the subsequent pregnancy (3 sessions across trimesters). Control participants will receive routine postpartum and antenatal care. Primary outcome is a composite indicator of maternal and neonatal morbidity in the subsequent pregnancy. Secondary outcomes will assess adherence, weight trajectories, and behavioral mediators. Data collection includes medical records, anthropometric measurements, device-generated activity and weight data, and questionnaires at standard intervals.